Unit II :: Real Gases

PHY0500304: Heat & Thermodynamics

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Syllabus

Unit II: Real Gases

Behavior of Real Gases: Deviations from the Ideal Gas Equation. The Virial Equation. Andrew's Experiments on CO2 Gas. Critical Constants. Continuity of Liquid and Gaseous State. Vapor and Gas. Boyle Temperature. Van der Waal's Equation of State for Real Gases. Values of Critical Constants. Law of Corresponding States. Comparison with Experimental Curves. Joule- Thomson Porous Plug Experiment. Joule- Thomson Effect, Joule-Kelvin coefficient for Ideal and Van der Waal Gases. Temperature of Inversion.

1 Deviations from the ideal gas equation

Real gases exhibit behaviour that departs from the predictions of the ideal gas equation, especially under conditions of high pressure and low temperature. These deviations arise mainly due to intermolecular forces and the finite size of gas molecules: two factors neglected in the ideal gas model. Understanding these departures is important because most practical gases behave non-ideally under ordinary laboratory conditions.

Real gases fail to obey the ideal gas law, $PV = nRT$, when intermolecular interactions and finite molecular volumes become significant. At high pressures and low temperatures, gases show noticeable departures from ideality, often represented using PV vs P or *compressibility factor* $Z = \frac{PV}{RT}$. These deviations provide important clues about the underlying physics governing real gas behaviour.

1.1 The virial equation

The virial equation provides a systematic, low-density expansion of a real gas's equation of state and quantifies departures from ideal behaviour in terms of temperature-dependent coefficients. Each virial coefficient encodes the contribution of clusters of a given size – two-body, three-body, etc. – so the series links macroscopic thermodynamics to microscopic intermolecular forces. It is most useful for dilute gases where truncation after the first few terms gives accurate results; it breaks down when density is high or near phase transitions.

We write the compressibility factor as

$$Z = \frac{PV_m}{RT}$$

where V_m is the molar volume.

Then the virial expansion in powers of the molar density $\rho \equiv \frac{1}{V_m}$ is

$$\boxed{Z = 1 + B(T)\rho + C(T)\rho^2 + D(T)\rho^3 + \cdots}$$

or equivalently, as an expansion in inverse molar volume:

$$\boxed{Z = 1 + \frac{B(T)}{V_m} + \frac{C(T)}{V_m^2} + \frac{D(T)}{V_m^3} + \cdots}$$

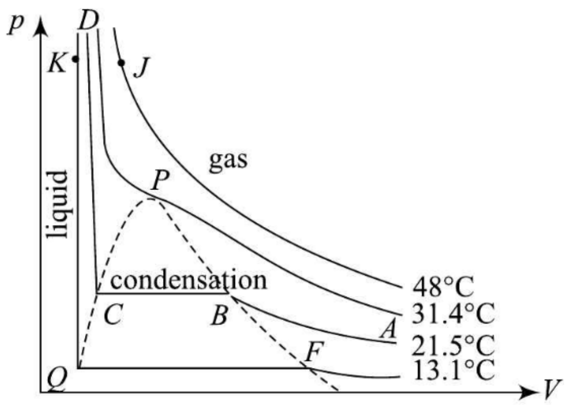
Here $B(T)$, $C(T)$, ... are the molar virial coefficients: B is the second virial coefficient (pair interactions), C the third (three-body effects), and so on.

1.2 Andrew's experiments on CO_2 gas

Andrew's experiments provided the first systematic evidence of the non-ideal behaviour of real gases. By studying the isotherms of CO_2, he demonstrated that gases do not liquefy merely by applying pressure unless their temperature is below a certain critical value. His work revealed the presence of characteristic flat portions of isotherms – signifying phase coexistence – which the ideal gas equation cannot predict. These experiments paved the way for the development of modern real-gas equations of state.

1.2.1 Description of Andrew's isotherms

For CO_2, Andrew plotted P-V isotherms at different temperatures.



- Above a certain temperature, the isotherms resemble smooth curves similar to ideal behaviour.
- Below that temperature, the isotherms flatten, indicating coexistence of liquid and gas.

The characteristic plateau corresponds to the liquid-vapour equilibrium region. Andrew also discovered the critical isotherm, at temperature T_c, where the flat portion disappears, and the gas cannot be liquefied by pressure alone.

1.3 Critical constants

Critical constants describe conditions under which the gas exhibits the *critical point* – a unique state where distinct liquid and gas phases cease to exist. At this point, the properties of the liquid and vapour become identical. These constants are essential for understanding gas–liquid transitions and for deriving real-gas equations. They are determined experimentally from P-V isotherms.

The three critical constants are:

1. *Critical temperature* T_c – the maximum temperature at which a gas can be liquefied by pressure alone.
2. *Critical pressure* P_c – the pressure required to liquefy a gas at T_c
3. *Critical volume* V_c – the molar volume at the critical point

At the critical point, the isotherm satisfies the following conditions:

$$\left(\frac{\partial P}{\partial V}\right)_{T_c} = 0 \quad \text{and} \quad \left(\frac{\partial^2 P}{\partial V^2}\right)_{T_c} = 0$$

1.4 Continuity of liquid and gaseous state

The concept of continuity emphasises that liquid and gaseous states are not fundamentally distinct but represent different regions of the same phase. As Andrew's isotherms show, one can move from a gas-like state to a liquid-like state without crossing a phase boundary – provided the temperature remains above the critical temperature. This insight marks a major departure from earlier ideas that regarded gases and liquids as entirely different. It also supports the modern view that phases are defined by thermodynamic properties, not by intrinsic physical differences.

Above T_c, continuous compression transforms a gaseous state smoothly into a highly compressed fluid without condensation. This region is often called the supercritical fluid region, where properties interpolate between liquid-like and gas-like characteristics.

1.5 Boyle temperature

The Boyle temperature represents the temperature at which a real gas shows minimal deviation from ideal behaviour over a moderate range of pressures. It serves as a useful reference for identifying temperatures at which intermolecular attractions and repulsions cancel out. At this temperature, the compressibility factor $Z \approx 1$ for a substantial pressure range. Thus the Boyle temperature is significant for gas approximation models.

The Boyle temperature T_B is defined using the compressibility factor expansion of the virial equation:

$$Z = 1 + B(T)\frac{1}{V_m} + C(T)\frac{1}{V_m^2} + \cdots$$

At Boyle temperature T_B, $B(T_B) = 0$ and so the second virial term vanishes. This means that, at the Boyle temperature, the pair-wise interactions do not occur anymore.

2 Van der Waal's equation of state for real gases

The van der Waals' equation introduces corrections to the ideal gas law to account for intermolecular attractions and finite molecular size. It is one of the earliest and most successful equations of state for real gases, providing a qualitative description of gas-liquid transitions. The equation refines our understanding of gas behaviour under non-ideal conditions and forms a foundation for advanced thermodynamic analysis.

The van der Waals' equation is:

$$\left(P+\frac{a}{V_m^2}\right)(V_m-b)=RT$$

where a accounts for attractive forces, b represents the effective molecular volume (excluded volume) and V_m is the molar volume of the gas.

2.1 Values of critical constants

The van der Waals' equation predicts explicit expressions for the critical constants. These values result from applying the critical point conditions to the equation of state. The critical constants allow a direct connection between the empirical critical parameters and the molecular constants a and b. This connection demonstrates the physical significance of the parameters.

At the critical temperature T_c, we have

$$\left(\frac{\partial P}{\partial V_m}\right)_{T_c}=0 \quad \text{and} \quad \left(\frac{\partial^2 P}{\partial V_m^2}\right)_{T_c}=0$$

Applying the above conditions on the van der Waal's equation, we can obtain the critical pressure P_c, critical volume V_c and critical temperature T_c to be

$$P_c=\frac{a}{27b^2}, \qquad V_c=3b \quad \text{and} \quad T_c=\frac{8a}{27Rb}$$

Then we get the corresponding compressibility factor to be

$$\begin{aligned} Z_c &= \frac{P_cV_c}{RT_c} \\ &= \frac{(a/27b^2)(3b)}{R(8a/27Rb)} \\ &= \frac{\cancel{a}}{\cancel{27}\cancel{b^2}}\cdot 3\cancel{b}\cdot\frac{\cancel{27}\cancel{R}\cancel{b}}{8\cancel{a}\cancel{R}} \\ \implies \frac{P_cV_c}{RT_c} &= \frac{3}{8} \end{aligned}$$

which is a universal constant for all van der Waals gases.

3 Law of corresponding states

The law of corresponding states states that all real gases, when compared at corresponding reduced conditions, exhibit similar behaviour. This principle implies that gas properties can be made universal by scaling them using critical constants. As a result, gases appear to obey a common reduced equation of state. This law plays a crucial role in studying complex gases and supercritical fluids.

The dimensionless reduced variables are defined as:

$$\begin{aligned} P_r = \frac{P}{P_c}, \qquad V_r = \frac{V}{V_c} \quad \text{and} \quad T_r = \frac{T}{T_c} \\ \implies P = P_r P_c, \qquad V = V_r V_c \quad \text{and} \quad T = T_r T_c \end{aligned}$$

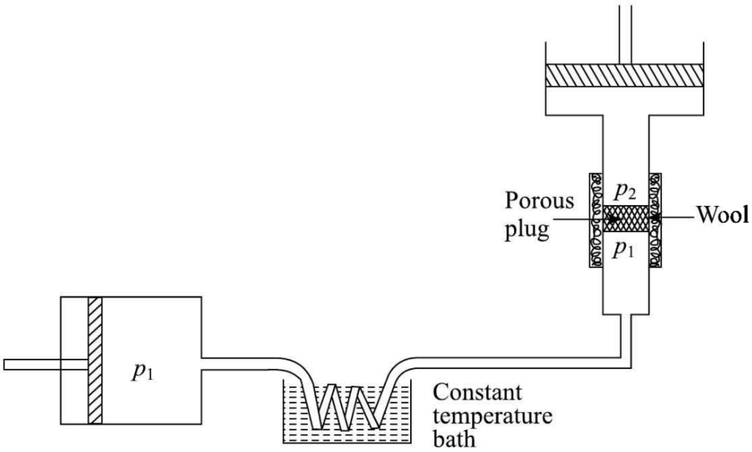
Then from the equation of state of a van der Waals' gas we obtain

$$\begin{aligned} \left(P + \frac{a}{V^2}\right)(V - b) &= RT \\ \implies \left(P_r P_c + \frac{a}{V_r^2 V_c^2}\right)(V_r V_c - b) &= R T_r T_c \\ \implies \left(P_r \frac{a}{27b^2} + \frac{a}{9b^2 V_r^2}\right)(3b V_r - b) &= \cancel{R} T_r \frac{8a}{27\cancel{R} b} \\ \implies \frac{a\cancel{b}}{9b^{\cancel{2}}}\left(\frac{P_r}{3} + \frac{1}{V_r^2}\right)(3V_r - 1) &= \frac{a}{27b} 8T_r \\ \implies \cancel{\frac{a}{27b}}\left(P_r + \frac{3}{V_r^2}\right)(3V_r - 1) &= \cancel{\frac{a}{27b}} 8T_r \\ \implies \boxed{\left(P_r + \frac{3}{V_r^2}\right)(3V_r - 1) = 8T_r} \end{aligned}$$

Thus gases with identical P_r, V_r and T_r behave similarly – thereby constituting the basis of corresponding states.

4 Joule-Thomson porous plug experiment

The Joule-Thomson porous plug experiment investigates how the temperature of a real gas changes when it expands through a porous barrier without performing external work. This setup models throttling processes used in refrigeration cycles. The experiment provides direct insight into the internal energy changes associated with real gas expansion. The observed temperature changes help define the Joule-Thomson coefficient.



In this experimental setup, a highly compressed gas is continuously forced at constant pressure to expand adiabatically and irreversibly through a porous plug kept in a non-conducting cylinder (as shown in the figure). The plug consists of a porous material, say cotton, wool, silk etc., and has a number of fine holes. Thus, a porous plus is equivalento a number of narrow orifices held in parallel.

The gas flows from a high-pressure region to a low-pressure region through a porous plug. Since the process is adiabatic and no external work is done, the enthalpy remains constant, i.e.

$$H_1 = H_2$$

This is known as *isenthalpic expansion*.

4.1 Joule-Thomson effect

The Joule-Thomson effect describes the temperature change experienced by a real gas during throttling at constant enthalpy. Depending on the nature of intermolecular forces, the gas may cool or warm upon expansion. The sign and magnitude of this effect are characterised by the Joule-Thomson coefficient. This behaviour is essential in refrigeration and liquefaction processes.

The Joule-Thomson coefficient is:

$$\boxed{\mu_{\text{JT}} = \left(\frac{\partial T}{\partial P}\right)_H}$$

- If $\mu_{\text{JT}} > 0$, the gas cools on expansion

- If $\mu_{\mathrm{JT}} < 0$, the warms up on expansion

The temperature at which $\mu_{\mathrm{JT}} = 0$ is known as the *inversion temperature*, above which expansion causes heating and below which cooling occurs.